



The Lowdown on Structural Insulated Panels

More builders are recognizing the benefits of building with structural insulated panels, yet unless the panels are properly installed, moisture buildup can create major problems.

by Al Cobb and Jim Gunshinan

Last summer's sharp spike in energy costs has caused an increased interest in the potential energy savings for homeowners and labor savings for homebuilders offered by structural insulated panel (SIP) technology (see "SIPs Face the Skeptics," *HE* Mar/Apr '98, p. 13). Now new test results from Oak Ridge National Laboratory (ORNL) confirm that SIPs can provide a higher whole-wall R-value than a comparable stick-built house.

Whole-wall measurements take into consideration heat loss due to seams and thermal bridging through wall studs, and are therefore more accurate than testing only the insulation material when measuring the R-values of buildings (see "Calculating Whole-Wall R-Values on the Net," *HE* Nov/Dec '99, p. 22). The 4-inch SIP scored R-14 on the whole-

wall tests, compared to R-9.8 for a 2 x 4 wood frame wall. The results of whole-wall tests of 6-inch SIPs compared to 2 x 6 wood stud walls were similar. The SIP wall scored R-21.6, while the wood stud wall scored a whole-wall R-value of 13.7 (see Figure 1).

These results are not that surprising, since SIP-built houses have fewer seams and therefore tend to be more airtight than stick-built houses. Also, since the insulation exists between two load-bearing panels, there is less framing needed in SIP building and therefore less thermal bridging through wall studs.

Jan Kosny, the author of the ORNL study, emphasizes that the SIPs must be installed correctly to avoid air and moisture infiltration. Laboratory conditions involve the best possible installation materials and techniques. Real practices,

if they are to achieve similar results, must be done just as well. Kosny adds that, with SIP building, there should be an emphasis on mechanical ventilation. If the SIPs are installed correctly, you end up with a pretty tight house. There has to be a way for fresh air to enter the living spaces. Without sufficient ventilation, high indoor humidity can lead to moisture buildup in occupied SIP homes during the winter in cold climates. In one Alaskan development, 37 out of 38 homes built with SIP roofs are experiencing major moisture problems, most likely due to poor installation.

In the field, Habitat for Humanity homes built with SIPs were tested for overall energy efficiency. The testing was spearheaded by the Florida Solar Energy Center, with support from Sumter County Habitat for Humanity, the Structural Insu-

Whole-Wall R-Value Comparison

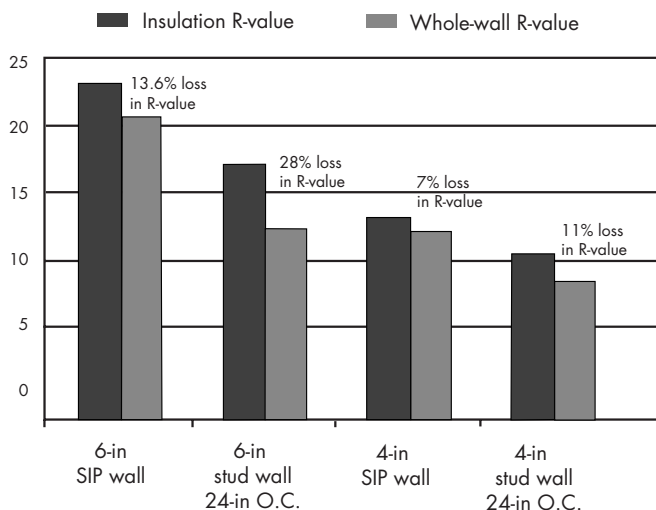


Figure 1. A comparison of SIP and stick-built whole-wall R-values shows that SIP walls provide more insulation than stick-built walls. **Photos (left)** SIP packages can significantly decrease the time needed to get a structure dried in.

Test results show remarkable differences in both categories. All three houses had HERS scores of 83%. The SIP house achieved a whole-house leakage rate of 1.8 ACH at 50 Pa compared to 3.9 ACH for the frame house. Monitoring equipment installed to measure the total energy use and heating energy use collected data for December 1998 and January 1999. The SIP houses used 25% less heating energy than the conventional house. Some of this difference was due to the fact that the stick-built house had ducts outside the conditioned space and the SIP houses had ducts within the conditioned space. But in a similar comparison study done in Louisville, Kentucky, where the stick-built and SIP houses both had ducts inside the conditioned space, the SIP house used 15% less heating energy than the stick-built house.

Builders Weigh In

Do these test results translate into actual energy savings in the real world? John Hensler, a builder with RJT

Homes, has become an SIP convert, as the energy savings his home buyers see have helped his sales flourish. In 1995 in Alpine, California, Hensler built Crown Hills, one of the first major developments to incorporate SIPs. According to data provided by San Diego Gas & Electric, the average cost of heating and cooling a 1,600 ft² home in Crown Hills was \$48.41 per month compared to \$125 per month for a similar-size stick-built home. Homeowners living in Crown Hills reported that not only were their utility bills half of what other Southern Californians were paying, but they lived in a much quieter house as well.

This feedback, combined with the speed with which the subdivision sold new homes, convinced RJT to use SIPs in their next development. The new development, Palmilla, located in La Quita, California, will boast 150 homes, all made with SIPs.

SIPs are proving their worth to many other builders because they save labor. The home builder is faced with not just a labor shortage but more specifically, a shortage of qualified labor. An SIP structure can be skinned with panels in less time than it takes to skin with conventional framing. With value-added features like complete precutting, SIP packages lessen the need for highly skilled carpenters. Full-service SIP suppliers routinely offer design and preengineering with complete precut packages. These packages significantly decrease the time needed to get a structure dried in. Decreased labor time can mean labor cost savings that will offset increased material costs. Mic Carmichael, a veteran panel installer from Nevada City, California, claims that "when comparing 2 x 6 stick-built walls with 3 1/2-inch core panels, the difference in cost is zero."

Need for Training

If all the results are proving that panels are an economical and energy-efficient building choice, then why isn't everyone using them? The main reason that SIPs still make up only about 2% of total housing starts mirrors the problems the truss industry faced 40 years ago. Lack of



lated Panel Association, (SIPA), and the U.S. Department of Energy. The testing was performed to provide comparison numbers for both airtightness and energy use for two houses built with SIPs and one built conventionally, with stick framing and batt insulation. All three houses were built on neighboring lots in Plains, Georgia, and were built to have a similar calculated energy performance, or HERS score.

knowledge in the building community severely cripples the manufacturers' ability to sell their products. SIP construction is truly a systems approach to building. From the preengineering to the details of fasteners and sealants, the builder merely needs to follow simple directions while using basic carpentry skills. The problem, however, is that these details are often ignored or not properly taught.

If the installers lack proper training or experience, the results can be alarming. When that all-important dew point is reached within a panel joint or connection, the condensation will begin to cause mold growth, if not full-scale rot. This rot can ultimately destroy the structural integrity of the system. Industry studies of SIPs with moisture problems have revealed in almost all cases a breakdown in proper installation techniques. SIP building is not the only construction that is susceptible to uncontrolled air migration. Any tight structure must be designed to control where and how air is exchanged. But if SIP building is to become a mainstream

building practice, SIP manufacturers and those who have experience with SIP building need to focus on proper training and installation.

Most of the training offered on SIP installation comes from the manufacturers in the form of short training seminars or on-site technical assistance. More extensive training is not readily available, and the Structural Insulated Panel Association (SIPA) recognizes this

shortcoming. With the assistance of industry experts, SIPA is working to develop a standard for panel installation details and techniques as well as a methodology for distributing SIP training throughout the country.



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For more information:

Kosny, J., A. Desjarlais, and J. Christian, *Whole Wall Rating/Label for Structural Insulated Panel: Steady-State Thermal Analysis*. Oak Ridge, Tennessee, Oak Ridge National Laboratory, Buildings Technology Center, June 1999.

For details on SIP building and to get in touch with manufacturers and installers of panel products, go to the SIPA Web site: www.sips.com.

For the comparison study done in Louisville, Kentucky, see Rudd, Armin, and Subrato Chandra. "Side-By-Side Evaluation Of a Stressed-Skin Insulated-Core Panel House and a Conventional Stud-Frame House," *Proceedings* from the 1994 EEBA Excellence Conference, 1994.

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